Truthful mechanisms for ownership transfer with expert advice

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Abstract

When a company undergoes a merger or transfer of its ownership, the existing governing body has an opinion on which buyer should take over as the new owner. Similar situations occur while assigning the host of big sports tournaments, e.g., the World Cup or the Olympics. In all these settings, the values of the *external* bidders are as important as the *internal* experts' opinions. Motivated by such questions, we consider a social welfare maximizing approach to design and analyze truthful mechanisms in these settings. We consider the problem with one expert and two bidders and provide *tight approximation guarantees* of the optimal social welfare. Since this problem is a combination of mechanism design with and without monetary transfers (payments can be imposed to the bidders but not to the expert), classical solutions like VCG cannot be applied, making this a unique mechanism design problem. We distinguish between mechanisms that use ordinal and cardinal information, as well as between mechanisms that base their decisions on one of the two sides (either the bidders or the expert) or both. Our analysis shows that the cardinal setting is quite rich and admits several non-trivial randomized truthful mechanisms, and also allows for closer-to-optimal-welfare guarantees.

1 Introduction

Most well-studied problems in *computational social choice* [Brandt *et al.*, 2016] deal with combining individual preferences over alternatives – often expressed as *rankings* – into a collective choice [Caragiannis *et al.*, 2017, Procaccia *et al.*, 2012, Skowron *et al.*, 2016]. More often than not, the mechanisms employed for this aggregation task are *ordinal*, i.e., they do not use the intensities of the preferences of the individuals, and *non-truthful*, which is justified by several impossibility theorems [Gibbard, 1973, 1977, Satterthwaite, 1975]. On the other hand, the class of truthful *cardinal* mechanisms has been shown to be much richer [Barbera *et al.*, 1998, Feige and Tennenholtz, 2010, Freixas, 1984] and the additional information provided by the numerical values can notably increase the well-being of society [Cheng, 2016, Filos-Ratsikas and Miltersen, 2014, Guo and Conitzer, 2010].

At the same time, truthful mechanisms *with money* are pretty well-understood and the welfare-maximizing mechanisms for a wide class of problems are known [Nisan *et al.*, 2007]. However, in a rich set of *hybrid social choice* problems, where monetary transfers are possible for some individuals and not for others, designing *truthful, cardinal* mechanisms is more challenging – one needs to combine elements of *mechanism design with money* and *social choice*.

Our interest in such settings emanates from several real-world scenarios. For instance, mergers and acquisitions play a central role in the competition among public or private organizations in a market,

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and the rules or the policies that dictate the mergers are often up for debate.[1] There is ample evidence to support the fact that the transfer of ownership of an organization has a significant impact on the economy of the employees and the customers [Auerbach, 2008, Hitt *et al.*, 2001]. Typically, in such transfers, the current governing body (e.g. the board of directors) nominates the potential new owner. Similarly, in the organization of sporting events, the bids of the potential hosts are taken into consideration along with the recommendations of the respective sports' administrative body (e.g., FIFA.com, 2018). Thus, our study applies to the problem of transferring the ownership of an organization or organizing rights of an event to a bidder with the advice of an expert in a *socially good manner*.

In this paper, we consider the setting where the bidders offer monetary compensations (e.g. they "buy-into" a new company), but the expert (e.g., the board of directors or the administrative body) does not. The objective is to achieve the decision that maximizes the *social welfare*, which includes the *cardinal* values of both the expert and the bidders. This is a hybrid social choice setting which is distinct from classical social choice, since the mechanism allows bidders to have monetary transfers, but also is distinct from the classical mechanisms with money, as the expert is never charged payments, thereby rendering celebrated solutions like the VCG mechanism [Clarke, 1971, Groves, 1973, Vickrey, 1961] insufficient.

1.1 Our contribution

We study the fundamental setting of two competing bidders A and B and one expert with cardinal preferences over the three options of selling to bidder A, selling to bidder B, or not selling at all (in which case the ownership transfer does not take place). We consider *truthful* mechanisms (i.e., mechanisms where neither a bidder nor the expert have any incentive to misreport their valuations) under different informational assumptions; *ordinal*, *bid-independent (cardinal)*, *expert-independent (cardinal)*, and general truthful mechanisms that take *both the bidders' and the expert's values* into account. We consider both *deterministic* and *randomized* mechanisms and use the notion of the *approximation ratio* to measure their quality in terms of social welfare. For all the classes defined by the above informational assumptions, we prove lower bounds on the approximation ratio of mechanisms in the class and identify the best possible among them. Our main results are summarized in Table 1.

class of mechanisms	approximation ratio	reference
ordinal	1.5	mechanisms EOM, BOM (Thm. 3)
		best possible (see Thm 4)
bid-independent	1.377	mechanism BIM (Thm. 7)
		best possible (Thm. 8)
expert-independent	1.343	mechanism EIM (Thm. 11)
		best possible (Thm. 11)
our template	1.25	randomized mechanism R (Thm. 14)
	1.618	deterministic mechanism D (Thm. 14)
		best possible, deterministic (Thm. 16)
all mechanisms	1.14	lower bound (Thm. 15)

Table 1: Overview of our results.

[1] EU data show that more than 6500 mergers have taken place in the EU since 1990, and strict rules are in effect for mergers [European Commission, accessed January 30 2018].

1.2 Related work

Our setting can be viewed as an instance of *approximate mechanism design*, *with* [Nisan and Ronen, 2001] and *without money* [Procaccia and Tennenholtz, 2013], which was proposed for problems where the goal is to optimize an objective under the strict truthfulness requirement. A result that will be very useful to our analysis is Myerson's characterization [Myerson, 1981] for single-parameter domains, which provides necessary and sufficient conditions for (deterministic or randomized) mechanisms (with money) to be truthful. On one hand, it allows us to abstract away from the payment functions (which are uniquely determined given the winning probabilities) on the bidders' side. On the other hand, similar arguments based on the same characterization enable us to reason about the structure of truthful mechanisms (without money) on the expert's side as well.

For settings with money, the well-known Vickrey-Clarke-Groves (VCG) mechanism [Clarke, 1971, Groves, 1973, Vickrey, 1961] is deterministic, truthful, and maximizes the social welfare in many settings of interest. As we pointed out in the introduction however, in our hybrid mechanism design setting, one needs to take the values of the expert into account as well and therefore the VCG mechanism is not longer optimal. Truthful mechanisms on the expert's side can be viewed as truthful voting rules; any positive result for deterministic such rules is impaired by the celebrated Gibbard-Satterthwaitte theorem [Gibbard, 1973, Satterthwaite, 1975] which limits this class to only *dictatorial mechanisms*.

However, on the contrary, the class of *randomized* truthful voting rules is much richer and includes many reasonable truthful rules that are not dictatorial. In fact, Gibbard [1977] characterized the class of all such *ordinal* rules; a general characterization of all *cardinal* voting rules is still elusive. To this end, a notable amount of work in the classical economics literature as well as in computer science has been devoted towards designing such rules and proving structural properties for restricted classes. Gibbard [1978] provided a similar characterization to his 1977 result, which however only holds for discrete strategy spaces, and later Hylland [1980][2] proved that the class of truthful rules that are Pareto-efficient reduces to random dictatorships. Freixas [1984], used the differential approach to mechanism design proposed by Laffont and Maskin [1980] to design a class of truthful mechanisms which actually characterize the class of twice differentiable mechanisms over subintervals of the valuation space; the best-possible truthful bid-independent mechanism that we propose in this paper can be seen as a mechanism in this class. Barbera *et al.* [1998] showed that there are many interesting truthful mechanisms that do not fall into the classes considered by Freixas [1984]. In the computer science literature, Feige and Tennenholtz [2010] designed a class of one-voter cardinal truthful mechanisms, where the election probabilities are given by certain polynomials.

The maximization of social welfare in settings without payments has been studied in a plethora of related papers in the computer science literature, in general social choice settings [Bhaskar *et al.*, 2018, Filos-Ratsikas and Miltersen, 2014], as well as in restricted domains,such as matching and allocation problems [Cheng, 2016, Filos-Ratsikas *et al.*, 2014, Guo and Conitzer, 2010]. Similarly to what we do here, Filos-Ratsikas and Miltersen [2014] make use of one-voter cardinal truthful mechanisms to achieve improved welfare guarantees. However, the presence of the bidders significantly differentiates our setting from theirs (as well as the other related works), since we have to consider both sides when calculating the approximation ratios and designing the mechanisms. Another relevant notion is that of the *distortion* of (non-truthful) mechanisms which operate under limited information (typically ordinal mechanisms) [Anshelevich *et al.*, 2015, Boutilier *et al.*, 2015, Caragiannis and Procaccia, 2011, Caragiannis *et al.*, 2016, 2017]. While the lack of information is also a restrictive factor for some of our results (in conjunction with truthfulness), we are mainly interested in cardinal mechanisms for which truthfulness is the limiting constraint.

[2]Quite remarkably, this paper is unpublished - the result was revisited in Dutta *et al.* [2007].

2 Preliminaries

Our setting consists of two agents A and B who compete for an item (to be thought of as an abstraction of a merger or acquisition) and an expert E. The agents have valuations w_A and w_B denoting the amount of money that they would be willing to spend for the item, and the expert has a valuation function $v : \mathcal{O} \to \mathbb{R}$ over the following three *options*: agent A is selected to get the item, or agent B is selected, or no agent is selected to get the item. We use $\oslash$ to denote this last option; hence, $\mathcal{O} = \{A, B, \oslash\}$. We use $\mathbf{w} = (w_A, w_B)$ to denote an *agent profile* and let $\mathcal{W}$ be the set of all such profiles. Similarly, we use $\mathbf{v} = (v(A), v(B), v(\oslash))$ to denote an *expert profile* and let $\mathcal{V}$ be the set of all such profiles. The domain of our setting is $\mathcal{D} = \mathcal{V} \times \mathcal{W}$. From now on, we use the term *profile* to refer to elements of $\mathcal{D}$.

A mechanism M takes as input a profile $(\mathbf{v}, \mathbf{w})$ and decides, according to a probability distribution (or *lottery*) P^M the pair $(o, \mathbf{p})$ consisting of an option $o \in \mathcal{O}$ and a vector $\mathbf{p} = (p_A, p_B)$ indicating the payments that are imposed to the agents. The execution of the mechanism yields a utility to the expert and the agents. Given an outcome $(o, \mathbf{p})$ of the mechanism, the utility of the expert is $u_E(o, \mathbf{p}) = v(o)$; the utility of agent i is $u_i(o, \mathbf{p}) = w_i - p_i$ if $i = o$ and $u_i(o, \mathbf{p}) = -p_i$ otherwise.

The expert and the agents are asked to submit an *expert's report* and *bids* to the mechanism and may have incentives to misreport their true values in order to maximize their utility. We are interested in mechanisms that do not allow for such strategic manipulations. We say that a mechanism M is *truthful for agent* i if for any agent value w_i and any profile $(\mathbf{v}', \mathbf{w}')$, $\mathbb{E}[u_i(M(\mathbf{v}', (w_i, w'_{-i}))] \geqslant \mathbb{E}[u_i(M(\mathbf{v}', \mathbf{w}'))]$, where the expectation is taken with respect to the lottery P^M. This means that bidding her true value w_i is a utility-maximizing strategy for the agent, no matter what the other agent and the expert's report are. Mechanism M is *truthful for the expert* if for any expert profile $\mathbf{v}$ and any profile $(\mathbf{v}', \mathbf{w}')$, $\mathbb{E}[u_E(M(\mathbf{v}, \mathbf{w}'))] \geqslant \mathbb{E}[u_E(M(\mathbf{v}', \mathbf{w}'))]$. Again, this means that reporting her true valuation profile is a utility-maximizing strategy for the expert, no matter what the agents bid. A mechanism M is *truthful* if it is truthful for the agents and truthful for the expert.

Our goal is to design truthful mechanisms that achieve high social welfare, which is the total value of the agents and the expert for the outcome. For a meaningful definition of the social welfare that weighs equally the expert's and the agents' valuations, we adopt a canonical representation of profiles. The expert has *von Neumann-Morgenstern valuations*, i.e., she has valuations of 0 and 1 for two of the options and a value in $[0, 1]$ for the third one. The agent values are normalized in the definition of the social welfare as follows. The social welfare of option $\oslash$ in a profile $(\mathbf{v}, \mathbf{w})$ is simply $\mathrm{SW}(\oslash, \mathbf{v}, \mathbf{w}) = v(\oslash)$. The social welfare of option $i \in \{A, B\}$ is $\mathrm{SW}(i, \mathbf{v}, \mathbf{w}) = v(i) + \frac{w_i}{\max\{w_A, w_B\}}$. We measure the quality of a truthful mechanism M by its *approximation ratio*, which (by abusing notation a bit and interpreting $M(\mathbf{v}, \mathbf{w})$ as the option decided by the mechanism) is defined as

$$\rho(M) = \sup_{(\mathbf{v},\mathbf{w}) \in \mathcal{D}} \frac{\max_{o \in \mathcal{O}} \mathrm{SW}(o, \mathbf{v}, \mathbf{w})}{\mathbb{E}[\mathrm{SW}(M(\mathbf{v}, \mathbf{w}), \mathbf{v}, \mathbf{w})]}.$$

Low values of $\rho(M)$, as close as possible to 1, are most desirable.

2.1 An alternative view of profiles

In order to simplify the exposition in the following sections, we devote some space here to introduce two alternative ways of representing profiles, which in turn will showcase more intuitive ways of realizing truthfulness and will help us in the design of efficient mechanisms.

Without essentially restricting the space of mechanisms that can achieve good approximation ratios according to our definition of the social welfare, we focus on mechanisms that base their decisions on the *normalized bid* values, i.e., on the quantities $\frac{w_A}{\max\{w_A, w_B\}}$ and $\frac{w_B}{\max\{w_A, w_B\}}$. It will be convenient to use the two alternative ways

$$\begin{pmatrix} 1 & x & 0 \\ h & \ell & z \end{pmatrix} \text{ and } \begin{bmatrix} h & \ell & n \\ 1 & y & 0 \end{bmatrix}$$

to represent the profiles, which is a different way of representing a profile $(\mathbf{v}, \mathbf{w})$. These representations are the expert's and agents' *view* of the profile, respectively. Each column corresponds to an option. According to the expert's view at the left, the columns are ordered in terms of the expert's values, which appear in the first row. The quantities h, ℓ, and z hold the normalized agent bids for the corresponding option and 0 for option $\oslash$. According to the agents' view at the right, the columns are ordered in terms of the bids, which appear in the second row. The quantities h, ℓ, and n hold the expert valuations for the corresponding options. These representations yield a crisper way to argue about truthfulness for the expert and the agents in our main results.

Similarly, we use two different representations of the lottery P^M, depending on whether we represent profiles according to the expert's or the agents' view. From the expert's viewpoint, P^M is represented by three functions g^M, f^M, and η^M, which correspond to the probability of selecting the first, second, and third favourite option of the expert, respectively. Similarly, from the agents' viewpoint, P^M is represented by three functions d^M, c^M, and e^M, which correspond to the probability of selecting the agent with the highest bid (or *high-bidder*), the other agent (or *low-bidder*), or option $\oslash$.

Let us present a example. Consider a profile with expert valuations 1 for option $\oslash$, 0.3 for option A, and 0 for option B and normalized bids of 1 and 0.9 from bidders A and B, respectively. Consider a lottery which, for the particular profile, uses probabilities 0.4, 0.1, and 0.5 for options A, B, and $\oslash$, respectively. The expert's and agents' views of the profile are

$$\begin{pmatrix} 1 & 0.3 & 0 \\ 0 & 1 & 0.9 \end{pmatrix} \text{ and } \begin{bmatrix} 0.3 & 0 & 1 \\ 1 & 0.9 & 0 \end{bmatrix},$$

respectively. Functions g^M, f^M, and η^M are defined over the 4-tuple of arguments $(x, h, \ell, z) = (0.3, 0, 1, 0.9)$ following the expert's view and take values 0.5, 0.4, and 0.1, respectively. Similarly, functions d^M, c^M, and e^M are defined over the 4-tuple of arguments $(y, h, \ell, n) = (0.9, 0.3, 0, 1)$ following the agents' view and take values 0.4, 0.1, and 0.5, respectively.

In order to handle situations of equal values (e.g., equal bids), we adopt the convention to resolve ties using the fixed priority $A \succ B \succ \oslash$ in order to identify the high- and low-bidder as well as the highest and/or lowest expert valuation. For example, if the expert has valuations of 1 for options $\oslash$ and B, we interpret this as option B being her most favourite one. Similarly, agent A is always the high-bidder and agent B is the low-bidder when their bids are equal. This is used in the definition of our mechanisms only; lower bound arguments do not depend on such assumptions in order to be as general as possible.

2.2 Reasoning about truthfulness

Let us now explain the truthfulness requirements having these profile representations in mind. There are two different kinds of possible misreportings by the expert.

- First, she can attempt to make a "level change in the reported valuation" (or ECh, for short) by changing her second highest valuation without affecting the order of her valuations for the options.
- Second, she can attempt to make a "reported valuation swap" (or ESw, for short), i.e., change the order of her valuations for the options as well as the particular values.

For example, the profile $\begin{pmatrix} 1 & 0.6 & 0 \\ 0.9 & 0 & 1 \end{pmatrix}$ is the result of reported valuation swap by the expert who changes her valuations from 1, 0.3, and 0 to 0.6, 0, and 1 for options $\oslash$, A, and B, respectively.

There are also two different kinds of possible misreportings by each agent. The agent can attempt to make

- a "level change in the reported bid" (BCh) by changing her bid without affecting the order of bids or

- a "bid swap" (BSw) by changing both the bid order and the corresponding values.

For example, the profile $\begin{bmatrix} 0 & 0.3 & 1 \\ 1 & 0.25 & 0 \end{bmatrix}$ is the result of a bid swap deviation by the low-bidder, who increases her bid in the profile above to a new bid that is four times the bid of the other agent.

A truthful mechanism never incentivizes (i.e., it is *incentive compatible* with respect to) such misreportings. We use the terms ECh-IC, ESw-IC, BCh-IC, and BSw-IC to refer to incentive compatibility with respect to the misreporting attempts mentioned above. A *truthful* mechanism, therefore, satisfies all these IC conditions. We continue with important conditions that are necessary and sufficient for BCh-IC and ECh-IC. The next lemma is essentially the well-known characterization of [Myerson, 1981] for single-parameter domains, which uniquely determines the payments that should be imposed to the agents so that BCh-IC is guaranteed. This allows us to focus on the lottery part of mechanisms only and ignore payments.

Lemma 1 (Myerson, 1981)**.** *A mechanism M is BCh-IC if and only if the functions d^M and c^M are non-increasing and non-decreasing in terms of their first argument, respectively.*

We have a similar proof to that of Myerson [1981] for characterizing ECh-IC in our setting.

Lemma 2. *A mechanism M is ECh-IC if and only if the function f^M is non-decreasing in terms of its first argument and the function g^M satisfies*

$$g^M(x,h,\ell,z) = g^M(0,h,\ell,z) - xf^M(x,h,\ell,z) + \int_0^x f^M(t,h,\ell,z)\,\mathrm{d}t, \tag{1}$$

for every 4-tuple (x,h,ℓ,z) representing a profile as seen by the expert.

As a corollary, functions g^M and h^M are non-increasing in terms of the first argument.

Proof. To shorten notation, we use $\mathbf{b} = (h,\ell,z)$ as an abbreviation of the information in the second row of a profile in expert's view and $(x,\mathbf{b})$ as an abbreviation of (x,h,ℓ,z). Also, we drop M from notation (hence, $f(x,\mathbf{b})$ is used instead of $f^M(x,h,\ell,z)$) since it is clear from context. Due to ECh-IC, the expert has no incentive to attempt a level change of her utility for her second favourite option from x to x'. This means that

$$g(x,\mathbf{b}) + xf(x,\mathbf{b}) \geqslant g(x',\mathbf{b}) + xf(x',\mathbf{b}). \tag{2}$$

Similarly, she has no incentive to attempt a level change of her utility for her second favourite option from x' to x. This means that

$$g(x',\mathbf{b}) + x'f(x',\mathbf{b}) \geqslant g(x,\mathbf{b}) + x'f(x,\mathbf{b}). \tag{3}$$

By summing (2) and (3), we obtain that

$$(x-x')(f(x,\mathbf{b}) - f(x',\mathbf{b})) \geqslant 0,$$

which implies that f is non-decreasing in terms of its first argument.

To prove equation (1), we observe that inequality (2) yields

$$g(x,\mathbf{b}) + xf(x,\mathbf{b}) \geqslant g(x',\mathbf{b}) + x'f(x',\mathbf{b}) + (x-x')f(x',\mathbf{b}). \tag{4}$$

This means that function $g(x,\mathbf{b}) + xf(x,\mathbf{b})$ is convex with respect to its first argument and has f as its subgradient [Rockafellar, 2015]. Hence, from the standard results of convex analysis we get

$$g(x,\mathbf{b}) + xf(x,\mathbf{b}) = g(0,\mathbf{b}) + \int_0^x f(t,\mathbf{b})\,\mathrm{d}t,$$

which is equivalent to (1). □

3 Ordinal mechanisms

We will consider several classes of truthful mechanisms depending on the level of information that they use. Let us warm up with ordinal mechanisms, which do not use the exact values of the expert's report and the bids but only their relative order. It turns out that the best possible approximation ratio of such mechanisms is $3/2$ and is achieved by two symmetric mechanisms, one depending only on the ordinal information provided by the expert (expert-ordinal), while the other depends only on the relation between the bids (bid-ordinal).

The expert-ordinal mechanism EOM selects the expert's favourite and second best option with probabilities $2/3$ and $1/3$, respectively. Symmetrically, the bid-ordinal mechanism BOM selects the high- and low-bidder with probabilities $2/3$ and $1/3$, respectively.

Theorem 3. *Mechanisms EOM and BOM are truthful mechanisms that have approximation ratio at most* $3/2$.

Proof. EOM is clearly truthful for the agents since it ignores the bids. It is also clearly truthful for the expert since the probabilities of selecting the options follow the order of the expert's valuations for them. BOM is clearly truthful for the expert (since her input is ignored); truthfulness for the bidders follows by observing that the probability of selecting an agent is non-decreasing in terms of her bid.

We prove the approximation ratio for mechanism BOM only; the proof for EOM is completely symmetric. Consider the profile $\begin{bmatrix} h & \ell & n \\ 1 & y & 0 \end{bmatrix}$ in agents' view. We distinguish between two cases. If $1 + h \geqslant y + \ell$, the optimal welfare is $1 + h$ and the approximation ratio is

$$\frac{1+h}{\frac{2}{3}(1+h) + \frac{1}{3}(y+\ell)} \leqslant \frac{3}{2}$$

since $y + \ell \geqslant 0$. If $1 + h \leqslant y + \ell$, the optimal welfare is $y + \ell$ and the approximation ratio is

$$\frac{y+\ell}{\frac{2}{3}(1+h) + \frac{1}{3}(y+\ell)} = \frac{1}{\frac{2}{3}\frac{1+h}{y+\ell} + \frac{1}{3}} \leqslant \frac{3}{2}$$

since $\frac{1+h}{y+\ell} \geqslant \frac{1}{2}$. □

We conclude the section by showing that both EOM and BOM are optimal among all ordinal mechanisms.

Theorem 4. *The approximation ratio of any ordinal mechanism is at least* $3/2$.

Proof. Let $\epsilon \in (0, 1/2)$ and consider the following two profiles:

$$\begin{pmatrix} 1 & \epsilon & 0 \\ 0 & \epsilon & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 1-\epsilon & 0 \\ 0 & 1-\epsilon & 1 \end{pmatrix}.$$

Since the order of the expert utilities and the bids is the same in both profiles, an ordinal mechanism behaves identically in all these profiles for every $\epsilon \in (0, 1/2)$. Assume that such a mechanism selects the middle option with probability p. Then, the approximation ratio of this mechanism is at least the maximum between its approximation ratio for these two profiles. Considering all profiles for $\epsilon \in (0, 1/2)$, we get an approximation ratio of at least

$$\sup_{\epsilon \in (0,1/2)} \left\{ \frac{1}{1-p+2\epsilon p}, \frac{2(1-\epsilon)}{1-p+2(1-\epsilon)p} \right\} = \max \left\{ \frac{1}{1-p}, \frac{2}{1+p} \right\}.$$

This is minimized to $3/2$ for $p = 1/3$. □

4 Bid-independent mechanisms

In this section, we consider cardinal mechanisms but restrict our attention to ones that ignore the bids and base their decisions only on the expert's report. It is convenient to use the expert's view of profiles $\begin{pmatrix} 1 & x & 0 \\ h & \ell & z \end{pmatrix}$. Then, a *bid-independent* mechanism can be thought of as using univariate functions g^M, f^M, and η^M which indicate the probability of selecting the expert's first, second, and third favourite option when she has value $x \in [0,1]$ for the second favourite option. We drop M from notation since the mechanism will be clear from context. The next lemma provides sufficient and necessary conditions for bid-independent mechanisms with good approximation ratio.

Lemma 5. *Let M be a bid-independent mechanism that uses functions g, f and η. Then M has approximation ratio at most ρ if and only if the inequalities*

$$2g(x) + xf(x) \geqslant 2/\rho \tag{5}$$

$$g(x) + (1+x)f(x) \geqslant (1+x)/\rho \tag{6}$$

hold for every $x \in [0,1]$.

Proof. Consider the application of M on the profile $\begin{pmatrix} 1 & x & 0 \\ h & \ell & z \end{pmatrix}$. If $1+h \geqslant x+\ell$ the optimal welfare is $1+h$ and the approximation ratio is

$$\begin{aligned} \frac{1+h}{(1+h)g(x) + (x+\ell)f(x) + z\eta(x)} &\leqslant \frac{1+h}{(1+h)g(x) + (x+\ell)f(x)} \\ &\leqslant \frac{2}{2g(x) + xf(x)}. \end{aligned}$$

The first inequality follows since $z, \eta(x) \geqslant 0$ and the second inequality follows since the expression at the RHS is non-increasing in ℓ and non-decreasing in h. Then, the first inequality of the statement follows as a sufficient condition so that M has approximation ratio at most ρ. To see why it is also necessary, observe that the inequalities in the derivation above are tight for $h = 1$, $\ell = 0$, and $z = 0$.

If $1 + h \leqslant x + \ell$ the optimal welfare is $x + \ell$ and the approximation ratio is

$$\begin{aligned} \frac{x+\ell}{(1+h)g(x) + (x+\ell)f(x) + z\eta(x)} &\leqslant \frac{x+\ell}{(1+h)g(x) + (x+\ell)f(x)} \\ &\leqslant \frac{1+x}{g(x) + (1+x)f(x)}. \end{aligned}$$

The first inequality follows since $z, \eta(x) \geqslant 0$ and the second inequality follows since the expression at the RHS is non-increasing in ℓ and non-decreasing in h. Then, the second inequality of the statement follows as a sufficient condition so that M has approximation ratio at most ρ. To see why it is also necessary, observe that the two inequalities in the derivation above are tight for $h = 0$, $\ell = 1$, and $z = 0$. □

Truthfulness of bid-independent mechanisms in terms of the agents follows trivially (since the bids are ignored). In order to guarantee truthfulness from the expert's side, we will use the characterization of ECh-IC from Lemma 2 together with additional conditions that will guarantee ESw-IC. These are provided by the next lemma.

Lemma 6. *An ECh-IC bid-independent mechanism is truthful if and only if the functions g, f, and η it uses satisfy $g(x) \geqslant f(x')$ and $f(x) \geqslant \eta(x')$ for every pair $x, x' \in (0,1)$.*

Proof. We first show that the first condition is necessary. Assume that the first condition is violated, i.e., $f(x_1) > g(x_2)$ for two points $x_1, x_2 \in (0,1)$. If $x_1 > x_2$, by the monotonicity of g we have $g(x_1) \leqslant g(x_2)$ and $f(x_1) > g(x_1)$. Otherwise, by the monotonicity of f, we have $f(x_2) \geqslant f(x_1)$ and $f(x_2) > g(x_2)$. In any case, there must exist $x^* \in (0,1)$ such that $f(x^*) > g(x^*)$. Now consider the swap from expert valuation profile $(1, x^*, 0)$ to the profile $(x^*, 1, 0)$. The utility of the expert in the initial true profile is $g(x^*) + x^* f(x^*)$ while her utility at the new profile becomes $f(x^*) + x^* g(x^*)$, which is strictly higher.

Now, we show that the second condition is necessary. Again, assuming that the second condition is violated, we obtain that there is a point $x^* \in (0,1)$ such that $\eta(x^*) > f(x^*)$. Now, the swap from expert's valuation profile $(1, x^*, 0)$ to the profile $(1, 0, x^*)$ increases the utility of the expert from $g(x^*) + x^* f(x^*)$ to $g(x^*) + x^* \eta(x^*)$, which is again strictly higher.

In order to show that the condition is sufficient for ECh-IC, we need to consider five possible attempts for valuation swap by the expert.

Case 1. Consider the swap from the valuation profile $(1, x, 0)$ to the profile $(1, 0, x')$. The utility of the expert at the new profile is $g(x') + x\eta(x') \leqslant g(0) + \int_0^x f(t)\,\mathrm{d}t = g(x) + xf(x)$, where the inequality holds due to the fact that $\eta(x') \leqslant f(t)$, for every $t \in [0, x]$. Observe that the RHS of the derivation is the expert's utility at the initial true profile.

Case 2. Consider the swap from the valuation profile $(1, x, 0)$ to the profile $(x', 1, 0)$. The utility of the expert at the new profile is $f(x') + xg(x') \leqslant g(x') + xf(x') = g(x') + x'f(x') + (x - x')f(x') \leqslant g(x) + xf(x)$, which is her utility at the initial true profile. The first inequality follows by the condition $g(x') \geqslant f(x)$ of the lemma and the second one is due to the convexity of function $g(x) + xf(x)$. See also the proof of Lemma 2.

Case 3. Consider the swap from the valuation profile $(1, x, 0)$ to the profile $(x', 0, 1)$. The utility of the expert at the new profile is $f(x') + x\eta(x')$, which is at most $g(x) + xf(x)$ due to the conditions of the lemma.

Case 4. Consider the swap from the valuation profile $(1, x, 0)$ to the profile $(0, x', 1)$. The utility of the expert at the new profile is $\eta(x') + xf(x') \leqslant f(x) + xg(x) \leqslant g(x) + xf(x)$, which is her utility at the initial true profile.

Case 5. Consider the swap from the valuation profile $(1, x, 0)$ to the profile $(0, 1, x')$. The utility of the expert at the new profile is $\eta(x') + xg(x') \leqslant f(x') + xg(x')$ and the proof proceeds as in Case 2 above. □

We are now ready to propose our mechanism BIM. Let $\tau = -W\left(-\frac{1}{2e}\right)$, where W is the Lambert function, i.e., τ is the solution of the equation $2\tau = e^{\tau - 1}$. Mechanism BIM is defined as follows:

$$f(x) = \begin{cases} \frac{\tau}{1+3\tau}, & x \in [0, \tau] \\ \frac{1+\tau-2\tau e^{1-x}}{1+3\tau}, & x \in [\tau, 1] \end{cases}$$

$$g(x) = \begin{cases} \frac{1+\tau}{1+3\tau}, & x \in [0, \tau] \\ \frac{2\tau(1+x)e^{1-x}}{1+3\tau}, & x \in [\tau, 1] \end{cases}$$

$$\eta(x) = \begin{cases} \frac{\tau}{1+3\tau}, & x \in [0, \tau] \\ \frac{2\tau(1-xe^{1-x})}{1+3\tau}, & x \in [\tau, 1] \end{cases}$$

BIM is depicted in Figure 1. All functions are constant in $[0, \tau]$ and have (admittedly, counter-intuitive at first glance) exponential terms in $[\tau, 1]$. Interestingly, as we will show later, this is the unique best possible bid-independent truthful mechanism. Its properties are proved in the next statement.

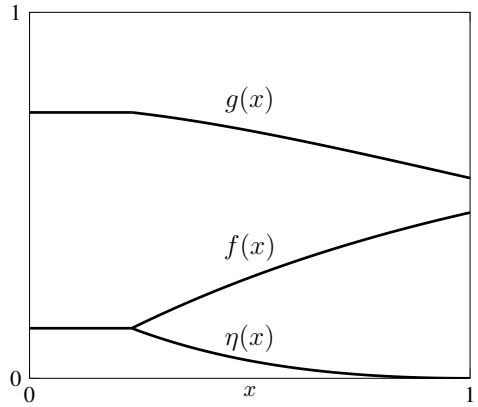

Figure 1: The lottery used by mechanism BIM.

Theorem 7. *Mechanism BIM is truthful and has approximation ratio at most* $\frac{1-3W\left(-\frac{1}{2e}\right)}{1-W\left(-\frac{1}{2e}\right)} \approx 1.37657$*, where* W *is the Lambert function.*

Proof. Tedious calculations can verify that BIM is truthful. The function f is non-decreasing in x and g is defined exactly as in equation (1); hence, ECh-IC follows by Lemma 2. ESw-IC follows since f, g, and h satisfy the conditions of Lemma 6.

Now, let $\rho = \frac{1+3\tau}{1+\tau}$. We use the definition of BIM and Lemma 5 to show the bound on the approximation ratio. If $x \in [0, \tau]$, inequalities (5) and (6) are clearly satisfied since $x \geqslant 0$ and $x \leqslant \tau$, respectively. If $x \in [\tau, 1]$, we have

$$2g(x) + xf(x) = 2\frac{2\alpha(1+x)e^{1-x}}{1+3\alpha} + x\frac{1+\alpha-2\alpha e^{1-x}}{1+3\alpha},$$

which is minimized for $x = \tau$ (recall that $\tau = e^{\tau-1}$) at $\frac{2+2\tau+\tau^2}{1+3\tau} \geqslant 2/\rho$. Hence, inequality (5) holds. Also, inequality (6) can be easily seen to hold with equality. □

We now show that the above mechanism is optimal among all bid-independent truthful mechanisms. The proof exploits the characterization of ECh-IC mechanisms from Lemma 2, the characterization of ESw-IC bid-independent mechanisms from Lemma 6, and Lemma 5.

Theorem 8. *The approximation ratio of any truthful bid-independent mechanism is at least* $\frac{1-3W\left(-\frac{1}{2e}\right)}{1-W\left(-\frac{1}{2e}\right)} \approx 1.37657$*, where* W *is the Lambert function.*

Proof. Let M be a bid-independent mechanism that uses functions g, f, and h to define the probability of selecting the expert's first, second, and third favourite option and has approximation ratio $\rho \geqslant 1$. By the necessary condition (1) for ECh-IC in Lemma 2, we know that

$$g(x) = g(0) - xf(x) + \int_0^x f(t)\,\mathrm{d}t. \tag{7}$$

Let α be any value in $[0, 1]$.

Due to the fact that $f(1) + g(1) \leqslant 1$, we have

$$g(0) + \int_0^1 f(t)\,\mathrm{d}t \leqslant 1. \tag{8}$$

By the necessary condition for ESw-IC in Lemma 6 and since g is non-increasing (by Lemma 2), we also have $f(x) \geqslant \eta(x) = 1 - f(x) - g(x) \geqslant 1 - f(x) - g(0)$, i.e., $g(0) + 2f(x) \geqslant 1$, for $x \in (0, 1)$. Integrating in the interval $(0, \alpha]$, we get

$$\alpha g(0) + 2\int_0^\alpha f(t)\,\mathrm{d}t \geqslant \alpha. \tag{9}$$

Since, the mechanism is ρ-approximate, Lemma 5 yields

$$g(0) \geqslant 1/\rho \tag{10}$$

(by applying inequality (5) with $x = 0$) and

$$g(x) + (1 + x)f(x) \geqslant (1 + x)/\rho, \forall x \in [\alpha, 1].$$

Using (7), this last inequality becomes

$$g(0) + f(x) + \int_0^x f(t)\,\mathrm{d}t \geqslant (1 + x)/\rho, \forall x \in [\alpha, 1].$$

Now, let λ be a continuous function with $\lambda(x) \leqslant f(x)$ in $[\alpha, 1]$ such that

$$g(0) + \int_0^\alpha f(t)\,\mathrm{d}t + \int_\alpha^x \lambda(t)dt + \lambda(x) = (1 + x)/\rho.$$

Setting $\Lambda(x) = \int_\alpha^x \lambda(t)dt$ (clearly, Λ is differentiable due to the continuity of λ in $[0, 1]$), we get the differential equation

$$g(0) + \int_0^\alpha f(t)\,\mathrm{d}t + \Lambda(x) + \Lambda'(x) = (1 + x)/\rho$$

which, given that $\Lambda(\alpha) = 0$, has the solution

$$\Lambda(x) = \frac{x}{\rho} - g(0) - \int_0^\alpha f(t)\,\mathrm{d}t + \left(g(0) - \frac{\alpha}{\rho} + \int_0^\alpha f(t)\,\mathrm{d}t\right)\exp(\alpha - x)$$

for $x \in [\alpha, 1]$. Hence,

$$\int_\alpha^1 f(t)\,\mathrm{d}t \geqslant \Lambda(1) = \frac{1 - \alpha e^{\alpha-1}}{\rho} - \left(1 - e^{\alpha-1}\right)g(0) - \left(1 - e^{\alpha-1}\right)\int_0^\alpha f(t)\,\mathrm{d}t. \tag{11}$$

Now, by multiplying inequalities (8), (9), (10), and (11) by coefficients 2, $e^{\alpha-1}$, $(2 - \alpha)e^{\alpha-1}$, and 2, respectively, and then summing them, we obtain

$$\rho \geqslant \frac{2 - \alpha e^{\alpha-1}}{2 - 3\alpha e^{\alpha-1} + 2e^{\alpha-1}}.$$

Picking $\alpha = -W\left(-\frac{1}{2e}\right)$ (i.e., α is the solution of the equation $e^{\alpha-1} = 2\alpha$), we get that

$$\rho \geqslant \frac{1 - 3W\left(-\frac{1}{2e}\right)}{1 - W\left(-\frac{1}{2e}\right)}.$$

This completes the proof. □

5 Expert-independent mechanisms

In this section, we consider mechanisms that depend only on the bids. Now, it is convenient to use the agents' view of profiles $\begin{bmatrix} h & \ell & n \\ 1 & y & 0 \end{bmatrix}$. Then, an expert-independent mechanism can be thought of as using univariate functions d^M, c^M, and e^M which indicate the probability of selecting the high-bidder, the low-bidder, and the option $\oslash$ in terms of the normalized low-bid y. Again, we drop M from notation. Following the same roadmap as in the previous section, the next lemma provides sufficient and necessary conditions for expert-independent mechanisms with good approximation ratio.

Lemma 9. *Let M be an expert-independent mechanism that uses functions d, c, and e with $d(y) = 1 - c(y)$ and $e(y) = 0$ for $y \in [0, 1]$. If*

$$\frac{1}{\rho} - \frac{1 - 1/\rho}{y} \leqslant c(y) \leqslant \frac{2(1 - 1/\rho)}{2 - y} \tag{12}$$

for every $y \in [0, 1]$, then M has approximation ratio at most ρ. Condition (12) is necessary for every ρ-approximate expert-independent mechanism.

Proof. Consider the application of M on the profile with agents' view $\begin{bmatrix} h & \ell & n \\ 1 & y & 0 \end{bmatrix}$. We distinguish between two cases. If $1 + h \geqslant y + \ell$, assuming that condition (12) is true, the approximation ratio of M is

$$\begin{aligned} \frac{1 + h}{(y + \ell)c(y) + (1 + h)(1 - c(y))} &= \frac{1}{\frac{y+\ell}{1+h}c(y) + 1 - c(y)} \\ &\leqslant \frac{1}{1 - (1 - y/2)c(y)} \leqslant \rho. \end{aligned}$$

The first inequality follows since $\frac{y+\ell}{1+h} \geqslant y/2$ when $y \in [0, 1]$, while the second one is essentially the right inequality in condition (12).

Otherwise, if $1 + h \leqslant y + \ell$, the approximation ratio of M is

$$\begin{aligned} \frac{y + \ell}{(y + \ell)c(y) + (1 + h)(1 - c(y))} &= \frac{1}{c(y) + \frac{1+h}{y+\ell}(1 - c(y))} \\ &\leqslant \frac{1 + y}{1 + yc(y)} \leqslant \rho. \end{aligned}$$

The first inequality follows since $\frac{1+h}{y+\ell} \geqslant \frac{1}{1+y}$ when $y \in [0, 1]$; again, the second one is essentially the left inequality in condition (12).

To see that condition (12) is necessary for every mechanism, first consider a mechanism M' that uses functions $\overline{c}$, $\overline{d}$, and $\overline{e}$ such that the function $\overline{c}$ violates the left inequality in (12), i.e., $\overline{c}(y^*) < \frac{1}{\rho} - \frac{1-1/\rho}{y^*}$ for some $y^* \in [0, 1]$. Then, using this inequality and the fact that $\overline{d}(y^*) \leqslant 1 - \overline{c}(y^*)$, the approximation ratio of M' at profile $\begin{bmatrix} 0 & 1 & 0 \\ 1 & y^* & 0 \end{bmatrix}$ is

$$\frac{y^* + 1}{(y^* + 1)\overline{c}(y^*) + \overline{d}(y^*)} \geqslant \frac{1 + y^*}{1 + y^*\overline{c}(y^*)} > \rho.$$

Now, assume that function $\overline{c}$ violates the right inequality in (12), i.e., $\overline{c}(y^*) > \frac{2(1-1/\rho)}{2-y^*}$. Then, using this inequality and the fact that $\overline{d}(y^*) \leqslant 1 - \overline{c}(y^*)$, the approximation ratio of M' at profile $\begin{bmatrix} 1 & 0 & 0 \\ 1 & y^* & 0 \end{bmatrix}$ is

$$\frac{2}{2\overline{d}(y^*) + y^*\overline{c}(y^*)} \geqslant \frac{2}{2 - (2 - y^*)\overline{c}(y^*)} > \rho$$

as desired. □

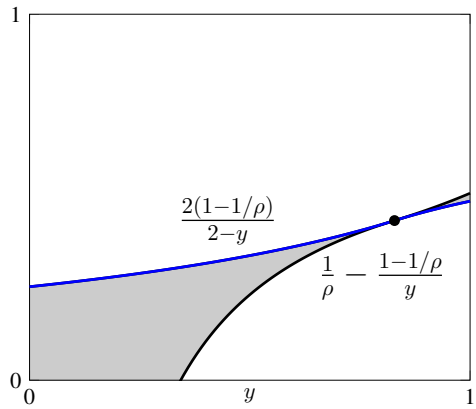

Figure 2: A pictorial view of the statement in Lemma 9.

Figure 2 shows the available space (grey area) for the definition of function $c(y)$, so that the corresponding mechanism has an approximation ratio of at most $\rho = 7 - 4\sqrt{2}$. It can be easily verified that this is the minimum value for which the LHS of condition (12) in Lemma 9 is smaller than or equal to the RHS so that a function satisfying (12) does exist.

Our aim now is to define an expert-independent truthful mechanism achieving the best possible approximation ratio of $\rho = 7 - 4\sqrt{2}$. Clearly, truthfulness for the expert follows trivially (since the expert's report is ignored). We restrict our attention to the design of a mechanism that never selects option $\oslash$, i.e., it has $d(y) = 1 - c(y)$ for every $y \in [0, 1]$. Lemmas 1 and 9 guide this design as follows. In order to be BCh-IC and ρ-approximate, our mechanism should use a non-decreasing function $c(y)$ in the space available by condition (12). Still, we need to guarantee BSw-IC; the next lemma gives us the additional sufficient (and necessary) condition.

Lemma 10. *A BCh-IC expert-independent mechanism is truthful if and only if* $d(1) \geqslant c(1)$.

Proof. Consider an attempted bid swap according to which the low-bidder increases her normalized bid of y so that it becomes the high-bidder and the normalized bid of the other agent is y'. Essentially, this attempted bid swap modifies the initial profile $\begin{bmatrix} h & \ell & n \\ 1 & y & 0 \end{bmatrix}$ to $\begin{bmatrix} \ell & h & n \\ 1 & y' & 0 \end{bmatrix}$. The deviating agent corresponds to the middle column in the initial profile and has probability $c(y)$ of being selected. In the new profile, she corresponds to the first column, and has probability $d(y')$ of being selected. So, the necessary and sufficient condition so that BSw-IC is guaranteed is $c(y) \leqslant d(y')$ for every $y, y' \in [0, 1]$. Since, by Lemma 1, c and d are non-decreasing and non-increasing, respectively, this condition boils down to $d(1) \geqslant c(1)$.

The case in which the high-bidder decreases her bid so that it gets a normalized value of y' is symmetric. □

We are ready to propose our mechanism EIM, which uses functions

$$c(y) = \begin{cases} \frac{2(1-1/\rho)}{2-y}, & y \in [0, \frac{3-\rho}{2}] \\ \frac{1}{\rho} - \frac{1-1/\rho}{y}, & y \in [\frac{3-\rho}{2}, 1] \end{cases}$$

for $\rho = 7 - 4\sqrt{2}$ and $d(y) = 1 - c(y)$ for $y \in [0, 1]$.

Essentially, EIM uses the blue line in the upper right plot of Figure 2, which consists of the curve that upper bounds the grey area up to point $\frac{3-\rho}{2} = 2\sqrt{2} - 2$ and the curve that lower-bounds the grey

area after that point. The properties of mechanism EIM are summarized in the next statement. It should be clear though that the statement holds for every mechanism that uses a non-decreasing function in the grey area that is below $1/2$ (together with the restriction $d(y) = 1 - c(y)$, this is necessary and sufficient for BSw-IC). Given the discussion about the optimality of $\rho = 7 - 4\sqrt{2}$ above, all these mechanisms are optimal within the class of expert-independent mechanisms.

Theorem 11. *Mechanism EIM is truthful and has approximation ratio at most* $7 - 4\sqrt{2} \approx 1.3431$. *This ratio is optimal among all truthful expert-independent mechanisms.*

6 Beyond expert-independent mechanisms

In this section, we present a template for the design of even better truthful mechanisms, compared to those presented in the previous sections. The template strengthens expert-independent mechanisms by exploiting a single additional bit of information that allows to distinguish between profiles that have the same (normalized) bid values.

We denote by $\mathcal{T}$ the set of mechanisms that are produced according to our template. In order to define a mechanism $M \in \mathcal{T}$, it is convenient to use the agents' view of a profile as $\begin{bmatrix} h & \ell & n \\ 1 & y & 0 \end{bmatrix}$. We partition the profiles of $\mathcal{D}$ into two categories. Category $T1$ contains all profiles with $\ell > h$ or with $\ell = h$ such that the tie between the expert valuations ℓ and h is resolved in favour of the low-bidder. All other profiles belong to category $T2$.

For each profile in category $T1$, mechanism M selects the low-bidder with probability $c(y, T1)$ that is non-decreasing in y and the high-bidder with probability $1 - c(y, T1)$. For each profile in category $T2$, mechanism M selects the low-bidder with probability 0, and the high-bidder with probability 1. Different mechanisms following our template are defined using different functions $c(y, T1)$. The mechanisms of the template ignore neither the bids nor the expert's report; still, it is not hard to show that they are truthful.

Lemma 12. *Every mechanism $M \in \mathcal{T}$ is truthful.*

Proof. We first show that M is truthful for the agents. BCh-IC follows easily by Lemma 1, since $c(y, T1)$ and $c(y, T2)$ are non-decreasing in y. To show BSw-IC, notice that a bid swap attempt from a profile of category $T1$ creates a profile of category $T2$ and vice versa. This involves either a high-bidder that decreases her bid and becomes the low-bidder in the new profile, or the low-bidder that increases her bid and becomes the high-bidder in the new profile. In both cases, the increase or decrease in the selection probability according to M follows the increase or decrease of the deviating bid.

To show that M is truthful for the expert, first observe that according to the expert's view, the lottery uses constant functions f, g, and h in terms of her valuation for her second favourite option. Hence, Lemma 2 implies ECh-IC. To show ESw-IC, observe again that an expert's report swap attempt from a profile of category $T1$ creates a profile of category $T2$ and vice versa. The expected utility that M yields to the expert in the initial profile is $\ell c(y, T1) + h(1 - c(y, T1)) = h + (\ell - h)c(y, T1) \geqslant h$ if it is of category $T1$ and $h + (\ell - h)c(y, T2) = h$ if it is of category $T2$. After the deviation, the utility of the expert becomes $\ell c(y, T1) + h(1 - c(y, T1)) = h + (\ell - h)c(y, T1) \leqslant h$ if the new profile is of category $T1$ and $h + (\ell - h)c(y, T2) = h$ if it is of category $T2$. Hence, such a swap attempt is never profitable for the expert. □

The next lemma is useful in proving bounds on the approximation ratio of mechanisms in $\mathcal{T}$.

Lemma 13. *Let M be a mechanism of $\mathcal{T}$ and $\rho \geqslant 1$ be such that the function $c(y, T1)$ used by M satisfies*

$$\frac{1}{\rho} - \frac{1 - 1/\rho}{y} \leqslant c(y, T1) \leqslant \frac{1 - 1/\rho}{1 - y}.$$

Then, M has approximation ratio at most ρ.

Proof. Clearly, the approximation ratio of M in profiles of category $T2$ is always 1 since the mechanism takes the optimal decision of selecting the high-bidder with probability 1.

Now, consider a profile $\begin{bmatrix} h & \ell & n \\ 1 & y & 0 \end{bmatrix}$ of category $T1$, i.e., $\ell \geqslant h$. We distinguish between two cases. If $1 + h \geqslant y + \ell$, then the approximation ratio of M is

$$\begin{aligned} \frac{1+h}{(y+\ell)c(y,T1) + (1+h)(1-c(y,T1))} &= \frac{1}{1 - c(y,T1) + \frac{y+\ell}{1+h}c(y,T1)} \\ &\leqslant \frac{1}{1-(1-y)c(y,T1)} \leqslant \rho. \end{aligned}$$

The first inequality follows since $\frac{y+\ell}{1+h} \geqslant y$ when $y \in [0,1]$ and $\ell \geqslant h \geqslant 0$ while the second one is due to the right inequality in the condition of the lemma.

Otherwise, if $1 + h \leqslant y + \ell$, the approximation ratio of M is

$$\begin{aligned} \frac{y+\ell}{(y+\ell)c(y,T1) + (1+h)(1-c(y,T1))} &= \frac{1}{c(y,T1) + \frac{1+h}{y+\ell}(1-c(y,T1))} \\ &\leqslant \frac{1+y}{1+yc(y,T1)} \leqslant \rho. \end{aligned}$$

The first inequality follows since $\frac{1+h}{y+\ell} \geqslant \frac{1}{1+y}$ when $y \in [0,1]$ and $h \geqslant \ell \geqslant 0$; again, the second one is due to the left inequality in the condition of the lemma. □

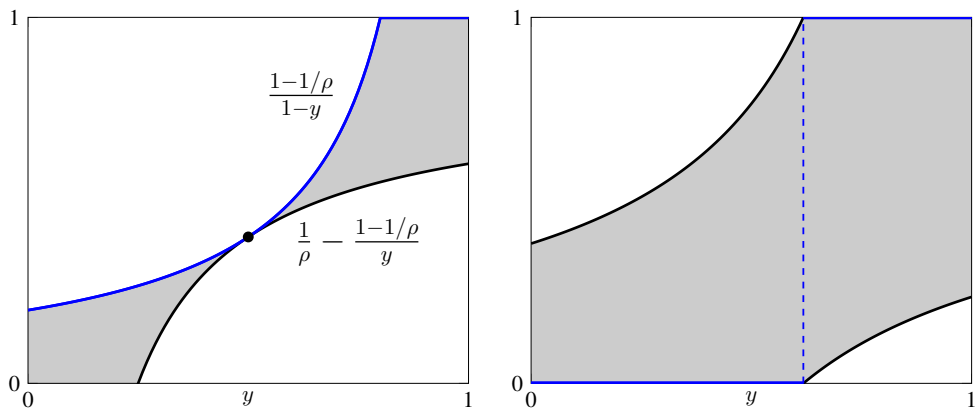

Figure 3: Pictorial views of the statement in Lemma 13.

The conditions in the statement of Lemma 13 are depicted in the left (for $\rho = 5/4$) and right plot (for $\rho = \phi$) of Figure 3. The grey area represents the available space for the definition of (the non-decreasing) function $c(y,T1)$ that a mechanism of $\mathcal{T}$ should use on profiles of category $T1$ so that its approximation ratio is at most ρ.

These plots explain the definition of the next two mechanisms that follow our template: the randomized mechanism R and the deterministic mechanism D. For each profile of category $T1$, mechanism R uses

$$c^R(y,T1) = \begin{cases} \frac{1}{5(1-y)}, & y \in [0,4/5] \\ 1, & y \in [4/5,1] \end{cases}$$

(i.e., the blue line in the lower left plot of Figure 2) and mechanism D uses

$$c^D(y, T1) = \begin{cases} 0, & y \in [0, 1/\phi] \\ 1, & y \in [1/\phi, 1] \end{cases}$$

(i.e., the blue line in the lower right plot of Figure 2), where $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$ is the golden ratio. Their properties are as follows.

Theorem 14. *Mechanisms R and D are $5/4$- and ϕ-approximate truthful mechanisms, respectively.*

Proof. Since $R, D \in \mathcal{T}$, their truthfulness follows by Lemma 12. The approximation guarantees follow by verifying that the conditions of Lemma 13 are satisfied for $\rho = 5/4$ and $\rho = \phi$, respectively. □

We remark that the condition of Lemma 13 can be proved to be not only sufficient but also necessary for achieving a ρ-approximation. Then, it can be easily seen that the value of $5/4$ is the lowest value for which the condition of the lemma is feasible. Hence, mechanism R is best possible among mechanisms that use our template. Mechanism D will be proved to be optimal among all deterministic truthful mechanisms in the next section.

7 Unconditional lower bounds

In the previous sections, we presented (or informally discussed) lower bounds on the approximation ratio of truthful mechanisms belonging to particular classes. Here, we present our most general lower bound that holds for every truthful mechanism. The proof exploits the ECh-IC characterization from Lemma 2.

Theorem 15. *The approximation ratio of any truthful mechanism is at least* 1.14078.

Proof. Let $\gamma \in [0, 1]$ be such that $1 - 2\gamma - 4\gamma^2 - 2\gamma^3 = 0$ and $\beta = (1 + \gamma)^{-1}$. The corresponding values are $\beta \approx 0.7709$ and $\gamma \approx 0.29716$.

Consider any ρ-approximate truthful mechanism and the profiles

$$\begin{pmatrix} 1 & \beta & 0 \\ \gamma & 1 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 0 & 0 \\ \gamma & 1 & 0 \end{pmatrix}.$$

Since the bids are identical in both profiles, we can assume that the functions f and g are univariate (depending only on the expert's second highest utility). Since the mechanism is ρ-approximate in both profiles, we have

$$(1 + \gamma)g(\beta) + (1 + \beta)f(\beta) \geqslant \frac{1 + \beta}{\rho} \tag{13}$$

and

$$(1 + \gamma)g(0) + f(0) \geqslant \frac{1 + \gamma}{\rho}. \tag{14}$$

By condition (1) in Lemma 2, $g(x) = g(0) - xf(x) + \int_0^x f(t)\,\mathrm{d}t$ which, due to the fact that f is non-decreasing (again by Lemma 2), yields $\int_0^\beta f(t)\,\mathrm{d}t \geqslant \beta f(0)$. Hence,

$$g(x) \geqslant g(0) - \beta f(\beta) + \beta f(0). \tag{15}$$

Also, clearly,

$$1 \geqslant g(\beta) + f(\beta). \tag{16}$$

Now, multiplying inequalities (13), (14), (15), and (16) by $\frac{\gamma}{\beta+2\beta\gamma-\gamma^2}$, $\frac{\beta-\gamma}{\beta+2\beta\gamma-\gamma^2}$, $\frac{(\beta-\gamma)(1+\gamma)}{\beta+2\beta\gamma-\gamma^2}$, and $\frac{\beta(1+\gamma)}{\beta+2\beta\gamma-\gamma^2}$, and by summing them, we get

$$\rho \geqslant \frac{\beta + 2\beta\gamma - \gamma^2}{\beta(1+\gamma)}.$$

Substituting β and γ, we obtain that $\rho \geqslant 1.14078$ as desired. □

Our last statement shows that mechanism D in Section 6 is best possible among all deterministic truthful mechanisms.

Theorem 16. *No truthful deterministic mechanism has approximation ratio better than ϕ.*

Proof. Let M be a deterministic truthful mechanism. Consider a profile $\begin{pmatrix} 1 & x & 0 \\ h & \ell & z \end{pmatrix}$ for some combination of values for h, ℓ, and z. We will first show that M selects the same option for every value of $x \in (0,1)$. Indeed, assume otherwise; due to Lemma 2, f must be non-decreasing in x and, hence, $f(x_1, h, \ell, z) = 0$ and $f(x_2, h, \ell, z) = 1$ for two different values x_1 and x_2 in $(0,1)$ with $x_1 < x_2$. Let $x_3 \in (x_2, 1)$, i.e., $f(x_3, h, \ell, z) = 1$ due to monotonicity. Property (1) in Lemma 2 requires that

$$g(x_3, h, \ell, z) = g(0, h, \ell, z) - x_3 + \int_0^{x_3} f(t, h, \ell, z)\,\mathrm{d}t.$$

By our assumptions on f (and due to its monotonicity), we also have that

$$x_3 - x_2 \leqslant \int_0^{x_3} f(t, h, \ell, z)\,\mathrm{d}t \leqslant x_3 - x_1.$$

These last two (in)equalities imply that $g(0, h, \ell, z) - g(x_3, h, \ell, z)$ lies between x_2 and x_3, i.e., it is non-integer. This contradicts the fact that M is deterministic.

Now let $\epsilon > 0$ be negligibly small and consider the two profiles

$$\begin{pmatrix} 1 & 1-\epsilon & 0 \\ 0 & 1/\phi & 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & \epsilon/\phi^2 & 0 \\ 0 & 1/\phi & 1 \end{pmatrix}.$$

If M selects the low-bidder in both profiles, its approximation ratio at the right one is $\frac{1}{\epsilon/\phi^2+1/\phi} \geqslant \phi - \epsilon$. Otherwise, its approximation ratio at the left profile is $1 + 1/\phi - \epsilon$. In any case, the approximation ratio is at least $\phi - \epsilon$, and the proof is complete. □

Of course, Theorem 16 is meaningful for cardinal mechanisms. Deterministic ordinal mechanisms can be easily seen to be at least 2-approximate.

8 Conclusion

We have presented a series of positive and negative results in a simple model which we believe that captures the main challenges in the implementation of ownership transfer. Still, closing the gap between the approximation ratio of $5/4$ achieved by mechanism R and our general lower bound of 1.14078 for any truthful mechanism seems elusive. Besides this concrete open problem, there are natural extensions of the model that are worth studying. For example, we have weighed equally the contribution of the

expert and the agents to the welfare. Generalizing this by introducing a factor of α, by which the contribution of the expert will be multiplied, is a first such extension. Of course, if α is very large or very small, bid-independent and expert-independent mechanisms will be almost optimal, respectively. But we suspect that there are values of parameter α close to 1 that make the mechanism design problem even more interesting. Of course, other generalizations include larger populations of experts and agents. Finally, the combination of mechanism design with and without money is definitely interesting in other contexts as well.

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